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MILITARY STANDARD

# PIEZOELECTRIC CERAMIC FOR SONAR TRANSDUCERS (HYDROPHONES AND PROJECTORS)

FSC 5845

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DEPARTMENT OF DEFENSE
Washington, DC 20362

Piezoelectric Ceramic for Sonar Transducers

DOD-STD-1376A(SH)(METRIC)

1. This Military Standard is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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# FOREWORD

1. This standard covers a system of designating piezoelectric ceramic materials by type, to define certain minimum and maximum properties for these types, and to aid in the choice of material selection. Part (A) should be referenced to define the type material required for hydrophones. Part (B) should be referenced to define the type material required for projectors.

2. The general properties specified permit a wide range of values for each type ceramic and are included in this standard solely for the purpose of defining the types of piezoelectric ceramic intended for use in Navy sonar transducers. Some of the properties and their associated values specified can only be determined by measurement on specific geometric shapes (for example, thin discs and rings) and that the values for these properties, such as the coupling factor and frequency constant, should not be specified in the acquisition of ceramic elements which do not meet the necessary geometric criteria. The individual specification for ceramic elements shall list required measurements and values, some of which can only be determined from actual measurements on the specific ceramic material and shape.



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## 1. SCOPE

1.1 Scope. This standard covers the types of piezoelectric ceramic materials required to manufacture sonar transducers for the Naval service. This standard covers the requirements for the ceramic compositions for four types based on standard test specimens and the procedure to certify piezoelectric ceramic manufacturers.

1.2 Classification. The standard ceramic types shall be of the following types:

Type I. Hard lead zirconate titanate with a Curie point equal to or greater than 310 degrees Celsius (°C).

Type II. Soft lead zirconate titanate with a Curie point equal to or greater than 330°C.

Type III. Very hard lead zirconate titanate with a Curie point equal to or greater than 290°C.

Type IV. Barium titanate with nominal additives of 5 percent calcium titanate and 0.5 percent cobalt carbonate with a Curie point equal to or greater than 100°C.

## 2. REFERENCED DOCUMENTS

2.1 Issue of documents. Not applicable.

2.2 Publications. The following publications form a part of this standard to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARD INSTITUTE (ANSI)
ANSI Y14.5 - Dimensioning and Tolerancing. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

ASTM
E 380 - Metric Practice. (DoD adopted)

(Application for copies should be addressed to the ASTM, 1916 Race Street, Philadelphia, PA 19103.)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)
IEEE 176 - Standard on Piezoelectricity.

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, 345 East 47th Street, New York, NY 10017.)



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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. DEFINITIONS

3.1 Terminology. The terminology used in this standard is based on definitions given in IEEE 176 except as follows:

3.1.1 Standard test specimens. The standard test specimens intended for verification of the material composition for hydrophones as defined in part (A) of this standard will be in the form of thin discs as specified in 5.1 through 5.3. The standard test specimens intended for verification of the material compositions for projectors as defined in part (B) of this standard will be in the form of rings as specified in 5.2 through 5.2.3.

3.1.2 Ceramic elements. Ceramic elements are those piezoelectric ceramic shapes (for example, bars, cylinders, discs, plates, rings, spheres and hemispheres) capable of performing for specific transducer requirements and are referred to as first article and production ceramic elements in this standard.

3.1.3 Individual equipment specifications. Drawings and specifications which describe ceramic elements shall be as specified in 3.1.2 and in accordance with the data ordering document (see appendix A).

3.2 Units and symbols. The International System of Units (SI) as shown in ASTM E 380 has been used where practical. A glossary of the symbols used in this standard is shown in table I.

3.3 Dimensioning and tolerancing. The dimensioning and tolerancing used to define the required condition of a part or component on an engineering drawing shall be in accordance with ANSI Y14.5.



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TABLE I. Glossary, symbols, definitions and units.

| Symbol | Definition | Unit |
|---|---|---|
| $C^T$ | Free capacitance (low frequency) | farad (F) |
| $D_m$ | Mean diameter | meter (m) |
| $\varepsilon_o$ | Permittivity of free space | $8.8542 \times 10^{-12}$ farads/meter |
| $\varepsilon_{33}^T$ | Free permittivity of material (low frequency) | farads/meter (F/m) |
| E | Applied electric field | kilovolt/meter (kV/m) |
| $f_m$ | Frequency of maximum admittance (minimum impedance) | hertz (Hz) |
| $f_n$ | Frequency of minimum admittance (maximum impedence) | hertz (Hz) |
| $K_{33}^T$ | Free relative dielectric constant = $\varepsilon_{33}^T/\varepsilon_o$ | |
| $k_{eff}^2$ | Effective coupling factor, $k_{eff}^2 = (f_n^2 - f_m^2)/f_n^2$ | |
| $k_p$ | Planar coupling factor (see figure 1) | |
| $\ell$ | Length | meter (m) |
| $N_1$ | Frequency constant for a ring, $N_1 = (f_m \pi D_m)/2$ | hertz-meter (Hz.m) |
| $N_p$ | Frequency constant planar mode disc, $N_p = (f_m.\text{Diam})$ | hertz-meter (Hz.m) |
| $Q_m$ | Mechanical quality factor, $Q_m = Y_m/(2\pi f_m C^T k^2_{eff})$ | |
| $\rho$ | Density | kilogram/meter$^3$ (kg/m$^3$) |
| $\sigma^E$ | Poisson's ratio, $\sigma^E = -s_{12}^E/s_{11}^E$ | |
| $s_{ij}^E$ | Elastic compliance at constant electric field | meter$^2$/newton (m$^2$/N) |
| $t$ | Thickness | meter (m) |
| tan $\delta$ | Dielectric loss factor | |
| $w$ | Width | meter (m) |
| $Y_m$ | Maximum admittance magnitude | siemens (S) (previously mhos) |
| $Y_n$ | Minimum admittance magnitude | siemens (S) (previously mhos) |



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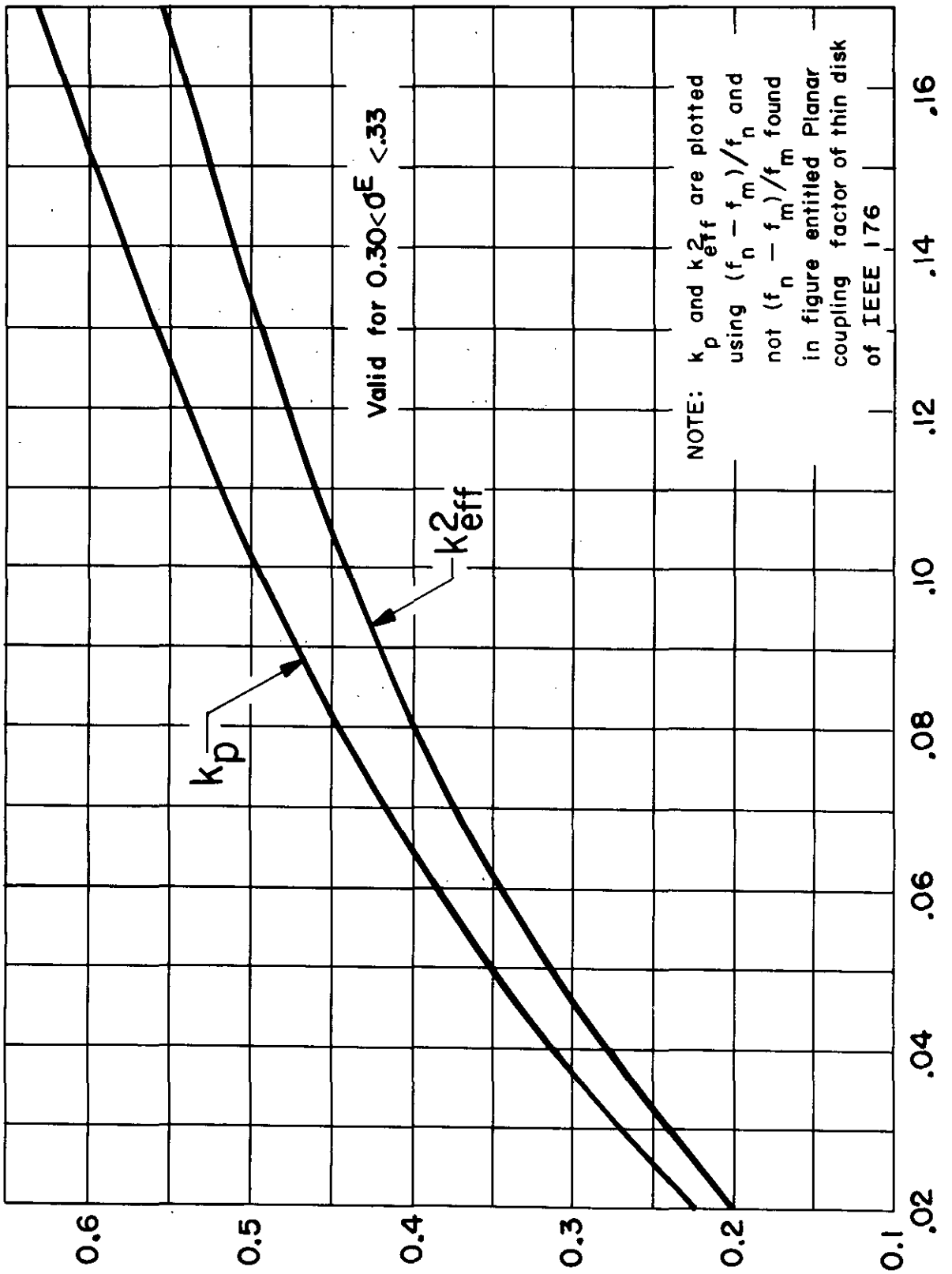



FIGURE 1. Planar and effective coupling factor of thin disc.



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## 4. GENERAL REQUIREMENTS

4.1 Standard ceramic types. The standard ceramic types shall be defined as follows:

(a) Type I. Hard lead zirconate titanate with a Curie point equal to or greater than 310°C.
(b) Type II. Soft lead zirconate titanate with a Curie point equal to or greater than 330°C.
(c) Type III. Very hard lead zirconate titanate with a Curie point equal to or greater than 290°C.
(d) Type IV. Barium titanate with nominal additives of 5 percent calcium titanate and 0.5 percent cobalt carbonate with a Curie point equal to or greater than 100°C.

NOTE: Standard test specimens shall be required only in acquisitions when a ceramic contractor must demonstrate the ability to produce the specific type(s) and is not currently certified as specified in 4.1.1.2.

4.1.1 Standard ceramic type compositions. The standard ceramic types to be used in sonar hydrophone receivers shall have the properties shown in table II. Types to be used in sonar projectors shall have the properties shown in table III and table IV. The properties shall be measured on standard test specimens as specified in part (A) and part (B). Lead zirconate titanate compositions shall have a nominal zirconate/titanate ratio of 53/47 modified with additives to meet the requirements of tables II, III or IV (where applicable), except as specified in 4.1.1.1.

4.1.1.1 Modified compositions. Deviations from the properties shown in tables II, III and IV for the standard ceramic types specified in 4.1 and 4.1.1 will be acceptable when required by the individual equipment specification.

4.1.1.2 Certification of manufacturers. A prospective piezoelectric ceramic manufacturer shall demonstrate the ability to meet the requirements of this standard for each type material to be manufactured as follows:

(a) Manufacture 10 or more standard test specimens of the specific type(s).
(b) Perform all tests specified herein for the particular application (hydrophone or projectors).
(c) Provide the test data to the contracting activity for examination and approval in accordance with the data ordering document (see appendix A).
(d) When required by the contracting activity, provide the standard test specimens to the designated testing laboratory for further verification.
(e) Obtain written certification of approval from the contracting activity.



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TABLE II. General properties and small signal values of hydrophone ceramic standard types measured with the standard disc.

| Property | | Material types | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | I | | | II | | | III | | | IV | | |
| | | 10-day value | | Aging 1/ rate percent | 10-day value | | Aging 1/ rate percent | 10-day value | | Aging 1/ rate percent | 10-day value | | Aging 1/ rate percent |
| | | Min | Max | | Min | Max | | Min | Max | | Min | Max | |
| Free relative dielectric constant | $K_{33}^T$ | 1100 | 1400 | -6 max -3 min | 1575 | 1875 | -1.5 max | 950 | 1150 | -5 max -2.0 min | 1100 | 1400 | -2 |
| Dielectric loss factor | tan $\delta$ | --- | 0.006 | --- | --- | 0.02 | --- | --- | 0.004 | --- | --- | 0.008 | --- |
| Piezoelectric coupling 2/ factor (planar) | $k_p$ | 0.52 | 0.60 | 6/ -2.5 | 0.54 | 0.62 | -0.3 | 0.47 | 0.55 | -2 | 0.28 | 0.33 | -2 |
| Frequency 3/ constant (Hz-m) | $N_p$ | 1990 | 2430 | 6/ +2.5 | 1795 | 2190 | +0.25 | 2035 | 2480 | +1.0 | 2870 | 3500 | +0.5 |
| Density $kg/m^3$ | $\rho$ | 7500 | --- | --- | 7600 | --- | --- | 7500 | --- | --- | 5550 | --- | --- |
| Mechanical 4/ 5/ quality factor | $Q_m$ | 500 | --- | --- | --- | --- | --- | 800 | --- | --- | 400 | --- | --- |
| Percentage 5/ change in $K_{33}^T$ 0 to 50°C | | --- | 10 | --- | --- | 30 | --- | --- | 10 | --- | --- | 6 | --- |

1/ The aging rate is the maximum and minimum (where specified) permitted change (percent) in properties in the period up to 100 days after poling using the 10 day value as the base.

2/ The planar coupling factor, $k_p$, will be determined from figure 1, where $f_n$ is the frequency of minimum admittance and $f_m$ is the frequency of maximum admittance; $f_m$ to be measured at fields $\leq$ 0.01 V/mm.

3/ Planar mode disc $N_p = (f_m \cdot \text{Diam})$.

4/ The mechanical quality factor, $Q_m$, defined by $Q_m = Y_m/(2\pi f_m C^T k_{eff}^2)$ where $C^T$ is the small signal capacitance measured at 1 kHz, $Y_m$ is the maximum admittance measured at $f_m$. Value for type II material is very low and is therefore not specified.

5/ Values for the mechanical quality factor and the change in $K_{33}^T$ with temperature are not 10-day values but shall be measured at approximately 100 days.

6/ Stated as typical value.



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TABLE III. General properties and small signal values of projector ceramic standard types measured with a standard ring.

| Property | | Material types | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | I | | | II[5/] | | | III | | | IV | | |
| | | 10-day value | | Aging[1/] rate percent | 10-day value | | Aging[1/] rate percent | 10-day value | | Aging[1/] rate percent | 10-day value | | Aging[1/] rate percent |
| | | Min | Max | | Min | Max | | Min | Max | | Min | Max | |
| Free relative dielectric constant | $K_{33}^T$ | 1100 | 1400 | -6 max -3 min | 1575 | 1875 | -1.5 | 950 | 1150 | -5 max -2.0 min | 1100 | 1400 | -2 |
| Dielectric loss factor | tan $\delta$ | --- | 0.006 | --- | --- | 0.02 | --- | --- | 0.004 | --- | --- | 0.008 | --- |
| Piezoelectric[2/] coupling factor (effective) | $k_{eff}^2$ | 0.31 | 0.36 | [7/] -2.5 | 0.32 | 0.38 | 0.3 | 0.27 | 0.33 | -2 | 0.145 | 0.20 | -2 |
| Frequency[3/] constant (Hz-m) | $N_1$ | 1500 | 1750 | [7/] +2.5 | 1300 | 1550 | +.25 | 1550 | 1880 | +1.0 | 2200 | 2500 | 0.5 |
| Density $kg/m^3$ | $\rho$ | 7500 | --- | --- | 7600 | --- | --- | 7500 | --- | --- | 5550 | --- | --- |
| Mechanical[4/6/] quality factor | $Q_m$ | 500 | --- | --- | --- | --- | --- | 800 | --- | --- | 400 | --- | --- |
| Percentage change[6/] in $K_{33}^T$ 0 to 50°C | | --- | 10 | --- | --- | 30 | --- | --- | 10 | --- | --- | 6 | --- |

1/ The aging rate is the maximum and minimum (where specified) permitted change (percent) in properties in the period up to 100 days after poling using the 10 day value as the base.

2/ The effective coupling factor is defined by $k_{eff}^2 = (f_n^2 - f_m^2)/f_n^2$ where $f_n$ is frequency of minimum admittance and $f_m$ is the frequency of maximum admittance; $f_m$ to be measured at fields $\leq$ 0.01 V/mm.

3/ The frequency constant $N_1$ for a thin ring is defined as $N_1 = (f_m \pi D_m)/2$, where $D_m$ is the mean diameter.

4/ The mechanical quality factor $Q_m$ defined as $Q_m = Y_m/(2\pi f_m C^T k_{eff}^2)$ where $C^T$ is the small signal capacitance measured at 1 kHz, $Y_m$ is the maximum admittance magnitude measured at $f_m$. Value for type II material is very low and is therefore not specified.

5/ Type II is not recommended for use in projector applications.

6/ Values for the mechanical quality factor and the change in $K_{33}^T$ with temperature are not 10-day values but shall be measured at approximately 100 days.

7/ Stated as typical value.





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TABLE IV. Large signal dielectric properties of projector ceramic standard types (measured in air).

| Property | | Material type | | | | |
|---|---|---|---|---|---|---|
| | | I | | III | IV | |
| Applied electric field kV/m (rms) | E | 200 | 400 | 400 | 150 | 300 |
| Max change in $K_{33}^T$ (percent) above small signal value (0.1 to 1.0 V/mm) | $\Delta K_{33}^T$ | 5 | 18 | 4.0 | 6 | 12 |
| Max dielectric loss factor | tan $\delta$ | 0.02 | 0.04 | 0.01 | 0.015 | 0.03 |

NOTE: Measurements at any frequency in the range 60 Hz to 2 kHz.

4.1.1.3 Prior certification. A ceramic contractor who has previously manufactured the specific ceramic types, shall submit objective evidence of prior test results and certification to the contracting activity. In the event the ceramic contractor has not manufactured the ceramic types during the 2 year period immediately preceding an acquisition, the contractor shall be required to be recertified in accordance with 4.1.1.2.

4.2 Requirements.

4.2.1 Standard test specimens. Conformance with 4.1 through 4.1.1.1 shall be determined by measurements on standard test specimens manufactured in accordance with 5.1 through 5.1.3 for discs and 5.2 through 5.2.3 for rings. The detailed requirements for these specimens are specified in 5.1 and 5.2, part (A) and part (B) respectively. Not less than 10 test specimens shall be manufactured for each ceramic type being tested. The requirements for first article and production ceramic elements shall be specified in the individual equipment specification.

4.2.2 Electrical measurements. The electrical measurements shall be made at a temperature between 20 and 25°C. Each test specimen shall be stored between these temperatures for at least 24 hours prior to the measurements. The temperature at which the measurements are made shall be recorded.

4.2.3 Workmanship. The standard test specimens shall meet all requirements specified herein. The specimens shall be clean and dry.

4.2.4 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the standard where such



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inspections are deemed necessary to assure that the test specimens conform to prescribed requirements. Test specimens manufactured for a ceramic type in performance of a contract for piezoelectric ceramic elements for the Government or a Government contractor shall be retained for a period of not less than 6 months after delivery of the ceramic elements unless specifically authorized in writing for earlier disposition by the cognizant Government inspector.

4.2.5 Inspection system requirements. The contractor shall provide and maintain an inspection system acceptable to the Government covering the supplies, fabricating methods and special tooling. The inspection system shall be in accordance with the data ordering document (see appendix A) for the standard test specimens.

5. DETAILED REQUIREMENTS

5.1 Part A - Hydrophone applications.

5.1.1 Dimensions and finish of standard test specimens. The standard test specimens for verification of the compositions for hydrophones shall be ceramic discs with a surface finish before electroding not to exceed 1.60 micrometers (63 microinch (μinch)). Each surface shall be flat within 0.05 millimeter (mm) (.002 inch) and the two flat surfaces of each disc shall be parallel within 0.05 mm (.002 inch). The dimensions of the finished poled discs shall be as follows.

| | |
|---|---|
| Diameter | 31.8 ± 0.15 mm (1.25 ± 0.006 inches) |
| Thickness | 2.5 ± 0.13 mm (0.10 ± 0.005 inch) |

5.1.1.1 Electrodes. Fired silver or electroless nickel electrodes shall be applied to the flat surfaces of the disc.

5.1.1.1.1 Adherence. The electrode adherence shall meet the test specified in 5.1.4.1.7.

5.1.2 Markings. The electrode to which the positive side of the power supply is attached during poling shall be marked as the positive with a suitable designation, such as a dot or +, with a nonconducting ink. The disc shall be marked to identify the manufacturer, the powder lot number, and the poling date. The markings shall be as small as practicable, but legible.

5.1.3 Poling. The standard test disc shall be poled through the thickness.

5.1.4 Requirements.

5.1.4.1 Examination and tests for standard test disc specimens.

5.1.4.1.1 Method. The measurements of 5.1.4.1.2 through 5.1.4.1.7 shall be performed in accordance with the methods of IEEE 176.



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5.1.4.1.2 Density. The density, $\rho$, of each test specimen shall be determined by weight and dimensions or the Archimedes Method. The measured value shall be equal to or greater than the value shown in table II for the particular type of material required. The density shall be determined to an accuracy of 0.2 percent.

5.1.4.1.3 Small signal properties. The small signal values of the free relative dielectric constant, $K^T_{33}$, the dielectric loss factor, tan $\delta$, the planar coupling factor, $k_p$, and the planar mode frequency constant, $N_p$, of each test specimen shall be measured three times in the period between 10 and 100 days after poling. $K^T_{33}$ and tan $\delta$ shall be measured at 1 kilohertz (kHz).

5.1.4.1.3.1 Time of measurement. The first set of measurements of $K^T_{33}$, tan $\delta$, $k_p$ and $N_p$ shall be made within 20 days after the poling, the second set approximately 45 days after poling, and the third set approximately 100 days after poling. The three sets of data shall be plotted against the logarithm of the time in days since poling and the values at 10 and 100 days shall be determined by extrapolation using a best fit straight line. The 10-day values and the aging rates (excluding the rate for tan $\delta$) for each time decade from 10 to 100 days shall fall within the limits of table II.

5.1.4.1.4 Quality factor, $Q_m$, and temperature characteristics of $K^T_{33}$. After the final set of small signal measurements of $K^T_{33}$, tan $\delta$, $k_p$ and $N_p$ have been completed, but not later than 110 days after poling, the mechanical quality factor, $Q_m$, and the temperature characteristics of $K^T_{33}$ shall be measured in that order and the corresponding values shall fall within the limits of table II. The percentage change in $K^T_{33}$ shall be determined during a heating cycle. The specimen shall be cooled to 0°C and held at this temperature for 1 hour with the electrodes shorted. Capacitance measurements shall be made at 0°C. The specimen, with electrodes shorted, shall be heated to 50°C and stabilized at this temperature for 1 hour and then capacitance measurements shall be made at this temperature. The percentage change shall be determined from these values using the 0°C value as the base.

5.1.4.1.5 Dimensions. The dimensions of each test specimen shall be determined by use of micrometers, calipers, surface flats and gages with certified accuracy to determine conformance with 5.1.

5.1.4.1.6 Electrode resistance. The resistance between any two points (and all points), on each electroded surface shall be less than 1.0 ohm for silver electrodes and less than 3.0 ohms for electroless nickel electrodes.

5.1.4.1.7 Electrode adherence. One electroded surface of all test specimens shall be cleaned with a solvent such as trichloroethane. After the surface is dry, a strip of pressure-sensitive adhesive at least 25 mm (nominal 1 inch) wide and 70 mm (2.76 inches) long with an adhesion greater than 50 ounce/inch (oz/in) 13.9 newton (N/25 mm) width, shall be pressed firmly across the diameter of the electrode. The tape shall remain in contact with the surface for at least 24 hours at a temperature between 20 and 25°C. One end of the tape shall then be lifted normal to the electroded surface forming a 90 degree angle. Continue to pull the tape until it is removed. The removal of any



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portion of the electroded surface with the tape shall be considered a failure to pass this test. More than two such failures out of ten test specimens is unacceptable. (Note: This test applies only to the standard test specimens. The test required for first article and production ceramic elements shall be specified in the individual equipment or ceramic element specification.)

5.2 Part B - Sonar projector application.

5.2.1 Dimensions and finish of standard test specimens. The standard test specimens for verification of compositions for projectors shall be ceramic rings with a surface finish before electroding not to exceed 1.60 micrometer (63 μinches). The outside diameter wall and the inside diameter wall shall be perpendicular within .250 mm (.010 inch) to the electrode surfaces. The dimensions of the finished poled rings shall be as follows:

Outside diameter 63.50 ± 0.635 mm (2.5 ± 0.025 inches)
Length or height 12.70 ± 0.25 mm (0.5 ± 0.010 inch)
Wall thickness 6.35 ± 0.760 mm (0.25 ± 0.030 inch) except that the wall thickness of any individual specimen shall not vary more than 0.760 mm (0.030 inch) total.

5.2.1.1 Electrodes. Fired silver or electroless nickel electrodes shall be applied to the flat surfaces of the rings.

5.2.1.1.1 Adherence. The electrode adherence shall meet the test specified in 5.2.4.1.8.

5.2.2 Markings. The rings shall be marked on the outer cylindrical surface with nonconducting ink to identify the manufacturer, the powder lot number, and the poling date. The electrode to which the positive side of the power supply is attached during poling shall be marked with a suitable designation, such as a dot or +, with a nonconducting ink on the outer cylindrical surface adjacent to that electrode face. The markings shall be as small as practicable, but legible.

5.2.3 Poling. The standard test rings shall be poled in the axial direction.

5.2.4 Requirements.

5.2.4.1 Examination and tests for standard test ring specimens.

5.2.4.1.1 Method. The measurements as specified in 5.1.4.1.2 through 5.1.4.1.7 shall be performed in accordance with the methods of IEEE 176.

5.2.4.1.2 Density. The density, ρ, of each test specimen shall be determined by the Archimedes Method. The measured value shall be equal to or greater than the value shown in table III for the particular type of material required. The density shall be determined to an accuracy of 0.2 percent.



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5.2.4.1.3 Small signal properties. The small signal values of the free relative dielectric constant, $K_{33}^T$, the dielectric loss factor, tan $\delta$, the effective piezoelectric coupling factor, $k_{eff}^2$, and the frequency constant, $N_1$ of each test specimen shall be measured three times in the period between 10 and 100 days after poling. $K_{33}^T$ and tan $\delta$ shall be measured at 1 kHz.

5.2.4.1.3.1 Time of measurements. The first set of measurements of $K_{33}^T$, tan $\delta$, $k_{eff}^2$, and $N_1$ shall be made within 20 days after the poling, the second set approximately 45 days after poling, and the third set approximately 100 days after poling. The three sets of data shall be plotted against the logarithm of the time in days since poling and the values at 10 and 100 days shall be determined by extrapolation using a best fit to a straight line. The 10-day values and the aging rates (excluding the rate for tan $\delta$) per time decade from 10 to 100 days shall fall within the limits shown in table III.

5.2.4.1.4 Quality factor, $Q_m$, and temperature characteristic of $K_{33}^T$. After the final set of small signal measurements of $K_{33}^T$, tan $\delta$, $k_{eff}^2$, and $N_1$ have been completed, but not later than 110 days after poling, the mechanical quality factor, $Q_m$, and the temperature characteristic of $K_{33}^T$ shall be measured in that order and the corresponding values shall fall within the limits shown in table III. The percentage change in $K_{33}^T$ shall be determined during a heating cycle. The specimen shall be cooled at 0°C and held at this temperature for 1 hour with the electrodes shorted. Capacitance measurements shall be made at 0°C. The specimen, with electrodes shorted, shall be heated to 50°C and stabilized at this temperature for 1 hour and then capacitance measurements shall be made at this temperature. The percentage change shall be determined from these values using the 0°C value as the base.

5.2.4.1.5 High field measurements. After all other measurements are completed, the free relative dielectric constant, $K_{33}^T$, and the dielectric loss factor, tan $\delta$, of each test specimen of type I, or III, or IV shall be measured at electric fields specified in table IV for the appropriate type of material and the measured values shall be less than or equal to the values shown in table IV. Modified types (see 4.1.1.1) used in sonar projector applications shall meet the requirements for the specified type (I, III or IV) unless otherwise specified in the individual equipment specifications. $K_{33}^T$ and tan $\delta$ shall be measured after a dwell of at least 1 minute at each voltage level. The electric field shall be reduced to 0 for 2 minutes between successive measurements. For reliable results on larger parts, this test shall be performed with leads soldered directly to the parts.

5.2.4.1.6 Dimensions. The dimensions of each test specimen shall be determined by use of micrometers, calipers, surface flats and gages with certified accuracy to determine conformance with the requirements of 5.1.

5.2.4.1.7 Electrode resistance. The resistance between any two points (and all points) on the electroded surface shall be less than 1.0 ohm for silver electrodes and less than 3.0 ohms for electroless nickel electrodes.



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5.2.4.1.8 Electrode adherence. One electroded surface of all test specimens shall be cleaned with a solvent such as trichloroethane. After the surface is dry, a strip of pressure-sensitive adhesive tape at least 50 mm (nominal 2 inches) wide and 80 mm (3.15 inches) long with an adhesion greater than 50 oz/inch (13.9 N/25 mm) width, shall be pressed firmly across the diameter of the electrode. The tape shall remain in contact with the surface for at least 24 hours at a temperature between 20 and 25°C. One end of the tape shall then be lifted normal to the electroded surface forming a 90 degree angle. Continue to pull the tape until it is removed. The removal of any portion of the electroded surface with the tape shall be considered a failure to pass this test. More than two such failures out of ten test specimens is unacceptable. (This test applies only to the standard test specimens. The test required for first article and production ceramic elements shall be specified in the individual equipment or ceramic element specification.)

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# APPENDIX A

## DATA REQUIREMENTS

10. DATA

10.1 Data requirements. When this standard is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL), the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDRL incorporated into the contract. When the provisions of DAR 7-104.9 (n)(2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this standard is cited in the following paragraphs.

| Paragraph no. | Data requirements title | Applicable DID no. | Option |
|---|---|---|---|
| 3.1.3 and appendix B, 50.8 | Drawings, engineering and associated lists | DI-E-7031 | Level 3<br>Design activity designation - Contractor<br>Design activity drawing numbers - Contractor<br>Delivery of hard copy - Contracting activity |
| 4.1.1.2 | Certificate of compliance | DI-E-2121 | ---- |
| 4.2.5 | Inspection system program plan | DI-R-4803 | ---- |

(Data item decriptions related to this standard, and identified in section 6 will be approved and listed as such in DoD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

10.1.1 The data requirements of 10.1 and any task in the standard required to be performed to meet a data requirement may be waived by the contracting/acquisition activity upon certification by the offeror that identical data were submitted by the offeror and accepted by the Government under a previous contract for identical item acquired to this standard. This does not apply to specific data which may be required for each contract, regardless of whether an identical item has been supplied previously (for example, test reports).



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# APPENDIX B

## GUIDANCE IN SELECTION OF TYPE OF CERAMIC

10. GENERAL

10.1 Scope. This appendix furnishes additional background to be used for guidance and to assist in the selection of the types of ceramic to be used when piezoelectric ceramic is required in the individual equipment specification.

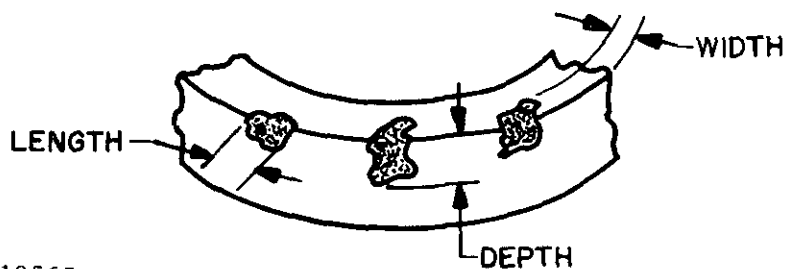
20. REFERENCED DOCUMENTS

20.1 Issue of documents. Not applicable.

30. DEFINITIONS

30.1 Definitions of physical flaws found in ceramic elements. The individual equipment specifications must specify the maximum size and number of each type that is acceptable in the ceramic element. In addition to any specific requirements, any flaw in the ceramic element large enough to degrade the performance for its intended use is unacceptable.

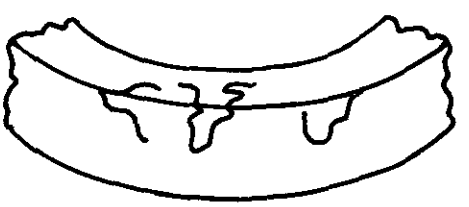
30.1.1 Open chips. Flaws at the intersection of two surfaces from which a fragment of the ceramic is missing (see figure 2).




SH 12565

FIGURE 2. Open chips.

30.1.2 Closed chips. Flaws in the ceramic consisting of chips that are not completely removed (see figure 3).



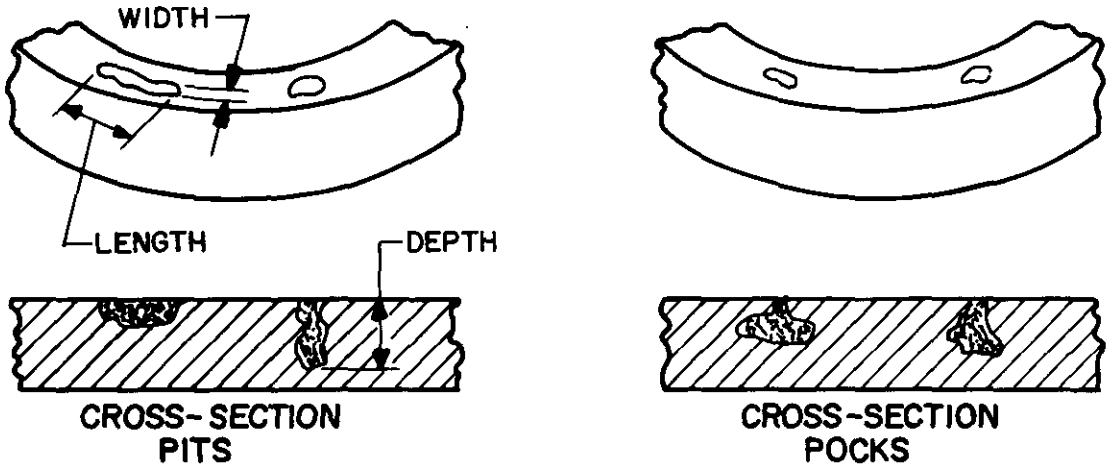
SH 12566

FIGURE 3. Closed chips.



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30.1.3 Pits and pocks. A pit is an open cavity on any surface. A pock is a partially closed pit (see figure 4).




SH 12567

FIGURE 4. Pits and pocks.

30.1.4 Cracks. A crack is a break or fissure on the surface or within the ceramic element which mechanically weakens the element and can cause it to part along the line of fracture.

40. GENERAL REQUIREMENTS

40.1 Manufacturing restrictions. Because of variations in raw materials, the piezoelectric ceramic general properties tend to drift or change over a time period. The manufacturer is dependent upon suppliers of lead oxide, titanium dioxide, zirconium dioxide, and so forth, with respect to purity, particle size, particle shape and particle size distribution. All of these parameters introduce variables which affect the characteristics of the final ceramic.

40.1.1 Despite these obstacles to a completely uniform product, most manufacturers of piezoelectric ceramic have developed techniques for adjusting the properties of their product within certain limits that can meet the requirements of most sonar element specifications.

40.1.2 The problem in meeting the specifications in production is essentially one of achieving the proper combination of dimensional requirements within the elastic, piezoelectric, and dielectric requirements and doing it economically.

40.2 Guidelines for specifications.

40.2.1 This standard is not intended to be used as a production specification for ceramic elements. This standard defines specific Navy piezoelectric ceramic types and provides the broad range of general properties which characterize that type. When specific electro-elastic properties are required, they must be specified in the individual equipment specifications and tolerances established for each parameter.



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40.2.2 When practicable, incorporate within the design a means of accommodating either changes in ceramic dimensions or electro-elastic properties.

40.2.3 Do not restrictively specify both dimensions and electro-elastic properties within too narrow limits.

40.2.4 If the design must be within narrow limits, determine which is the more economical, to accommodate adjustment in the electrical design, or to pay a higher price for closely controlled or selected ceramic elements.

40.2.5 Unless the transducer design uses idealized shapes, do not expect the nominal frequency constants and coupling factors in the standard to be applicable. When certain dimensions and dimensional ratios are used, they give rise to multiple mode couplings which can seriously alter the behavior of the ceramic element. Care must be exercised in selecting the dimensions of the ceramic element to minimize the excitation of multiple modes.

40.2.6 Prepare as complete specifications as possible to provide the manufacturer with realistic requirements and specify the most important physical and electro-mechanical properties.

40.2.7 This standard should be used as a guide in preparing the specification and to identify the type material to be specified. In hydrophone designs where uniformity of sensitivity is important, it may be necessary to establish limits on the piezoelectric constants g or d, for that particular application. In some cases, a method for measuring the sensitivity of the individual elements should be established, as for example, the use of a USRD type G19 bench calibrator or equivalent for measuring the sensitivity of capped cylinders.

40.2.8 On unusual design configurations or where close tolerances on either or both electrical and mechanical parameters are required, consult with the ceramic manufacturers to determine realistic tolerances that can be maintained in production quantities before the specifications are finalized.

50. DETAILED REQUIREMENTS

50.1 Characteristic frequencies. The characteristic frequencies which are usually required to evaluate the equivalent circuit parameters are the motional (series) resonance frequency, $f_s$, and the parallel resonance frequency, $f_p$. There are three pairs of frequencies of interest which coincide for a lossless circuit: that is, $f_m = f_r = f_s$ and $f_n = f_a = f_p$ where $f_r$ is the resonance frequency (susceptance = 0) and $f_a$ is the antiresonance frequency (susceptance = 0), and these occur at the minimum and maximum of impedance respectively. Since $f_s$ and $f_p$ are difficult to measure directly, $f_m$ and $f_n$ can be used in their place when the losses are small. A vector admittance diagram of a piezoelectric resonator is shown on figure 5. Additional information can be found in the references.



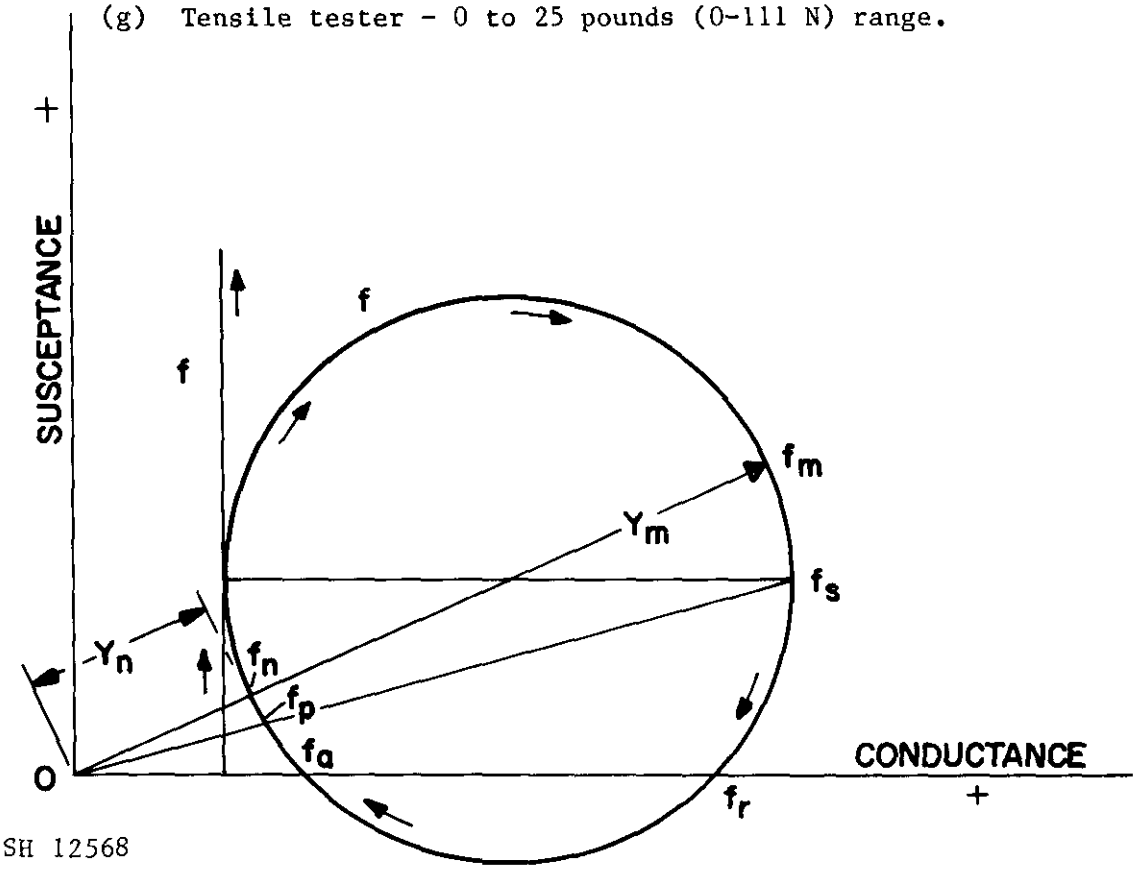
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50.2 Electrodes. The electrode material may be silver, electroless nickel, copper, electro-plated nickel or gold. Fired silver is the most common type found in the industry while electroless nickel has been used to a lesser extent. The quality of an electrode is best defined by the three factors: adhesion, surface finish and conductivity.

50.3 Electrode adhesion test. One method for testing for electrode adhesion is detailed hereinafter. Since this is a destructive test, the quantity of test specimens in a given lot or order must be specified.

50.3.1 Equipment required.

(a) Soldering iron - 40 watts (approximately).
(b) Fluxes: noncorrosive rosin flux for fired silver; Superior #30 (or equivalent) for electroless nickel.
(c) Tinned test wires 0.64 mm (0.025 inch) diameter solid wire with 1.3 mm (0.05 inch) diameter upset heads. (Available commercially such as resistor leads).
(d) Solder (silver electrodes) 62 percent tin, 36 percent lead and 2 percent silver.
(e) Solder (electroless nickel electrodes) 60 percent tin, 40 percent lead.
(f) A lead alinement fixture.
(g) Tensile tester - 0 to 25 pounds (0-111 N) range.




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FIGURE 5. Vector admittance diagram of a piezoelectric resonator.



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50.4 Preliminary preparations.

(a) Allow the soldering iron to preheat for at least 20 minutes before continuing.
(b) Thoroughly clean area to be soldered with an abrasive device (such as a pencil eraser).
(c) Flux both the ceramic electrode area to be soldered and the test wire.
(d) Pretin the electrode area and the test wire avoiding excessive flow-out of the solder on the electroded area.

50.5 Soldering procedure.

(a) Touch tinned tip of lead to surface of appropriate flux.
(b) Clip lead into alinement fixture with tinned end down.
(c) Position lead over the clean pretinned electrode surface to be evaluated and press down firmly onto the surface to permit flux wetting.
(d) Immediately touch soldering iron tip to the lead approximately 2 to 3 mm (0.125 inch) above the tinned end.
(e) While applying firm downward finger pressure, heat the lead until solder melts and wets the electrode surface.
(f) Excessive flowing out of solder on the electrode surface is to be avoided. (Not greater than 3.0 mm (0.125 inch diameter).
(g) As soon as solder melts, withdraw the soldering iron but do not allow any movement of the lead for at least 5 seconds to allow the solder to solidify into a firm joint.
(h) After soldered joint has solidified, open the alinement fixture clip and remove the fixture without touching the lead.
(i) It is not necessary to remove excess flux from the electrode surface.

50.6 Tensile testing.

(a) Set the tensile tester for the 0 to 25 pound (0 to 111 N) range and set the pointer to zero.
(b) Install the test sample, exercising care not to cause lateral bending of the lead.
(c) Exert tensile load on the lead at a rate of 5 to 10 mm/min (0.2 to 0.4 inch/min).
(d) Continue loading until the lead separates or an 8 pound (35 N) pull is indicated on the dial.
(e) Record the indicated value, relax and load, remove the lead from the fixture and repeat the test procedures for a total of six test leads (three for each major electroded surface unless otherwise specified). Not more than one pull under 8 pounds (35 N) is allowed.

NOTE: Exceptions to adhesion test: The adhesion test outlined in 50.3 through 50.6 is not applicable to ceramic parts having a dimension between the electrodes of 1.27 mm (.050 inch) or less. In addition, the adhesion test is not applicable where any electroded dimension is smaller than 1.57 mm (.062 inch). Devising a



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test for inside curved surfaces on cylinders, tubes or other shapes may not always be practicable. In some applications a pressure sensitive adhesive tape, with an adhesion of 50 to 100 oz/inch (14 to 28 N/25 mm) width, can provide a suitable test (see 5.4.1.7).

50.7 Mechanical dimensions and tolerances. Typical variations in mechanical dimensions and tolerances are shown in table V for cylinders, tubes and rings; and table VI for discs and plates. These tables can be used as a guide in preparing specifications; however, the ceramic manufacturer should be consulted to determine that a specific tolerance can be maintained on production quantities.

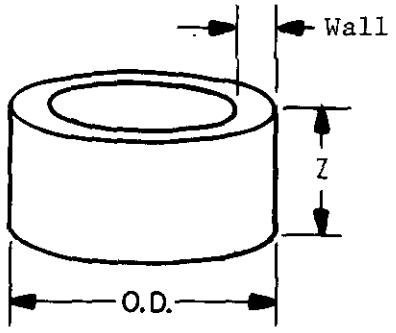
50.8 Ceramic element drawing. Figure 6 is a typical format which may be used in preparing the ceramic element drawing. It is not all inclusive. Other electrical parameters may be specified in place of or in addition to the parameters noted. The drawing should include requirements on the maximum size and number of open chips, closed chips or pits and pocks permitted. The drawing should be prepared in accordance with the data ordering document (see appendix A).

50.9 Suggested ordering data or specification guide. Table VII presents information which is typically required in specifying or describing a ceramic element. It is not necessary to specify every item listed. Only the items necessary to describe the element in sufficient detail to obtain the required performance for the intended purpose should be included. Over specifying the ceramic element will not only increase the cost but can invite diametrically opposite requirements.



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TABLE V. Mechanical tolerance levels for cylinders-tubes-rings.




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| Cylinder-tube-ring (o.d.) | | Level A Fully machined(±) | | Level B Minimal machining(±) | | Level C as fired (±) | |
|---|---|---|---|---|---|---|---|
| Inch | Millimeter | Inch | Millimeter | Inch | Millimeter | Inch | Millimeter |
| 0.250-0.500 | 6.35- 12.7 | 0.002 | 0.05 | 0.010 | 0.25 | 0.020 | 0.50 |
| .500-1.000 | 12.7 - 25.4 | .003 | 0.08 | .015 | 0.38 | .030 | 0.76 |
| 1.000-2.000 | 25.4 - 50.8 | .003 | 0.08 | .025 | 0.64 | .050 | 1.27 |
| 2.000-3.000 | 50.8 - 76.2 | .004 | 0.10 | .030 | 0.76 | .060 | 1.52 |
| 3.000-4.000 | 76.2 -101.6 | .004 | 0.10 | .050 | 1.27 | .090 | 2.29 |
| 4.000-6.000 | 101.6 -152.4 | .005 | 0.13 | .060 | 1.52 | .125 | 3.18 |
| (Wall) | | | | | | | |
| 0.020-0.031 | 0.51- 0.79 | .002 | 0.05 | 0.005 | 0.13 | 0.005 | 0.13 |
| .031- .063 | 0.79- 1.60 | .002 | 0.05 | .005 | 0.13 | .010 | 0.25 |
| .063- .100 | 1.60- 2.54 | .003 | 0.08 | .010 | 0.25 | .015 | 0.38 |
| .100- .125 | 2.54- 3.12 | .003 | 0.08 | .015 | 0.38 | .020 | 0.51 |
| .125- .250 | 3.17- 6.35 | .004 | 0.10 | .020 | 0.51 | .030 | 0.76 |
| .250- .350 | 6.35- 8.89 | .004 | 0.10 | .030 | 0.76 | .040 | 1.02 |
| .350- .500 | 8.89- 12.70 | .005 | 0.13 | .035 | 0.89 | .050 | 1.27 |
| (Length Z) | | | | | | | |
| 0.125-0.250 | 3.17- 6.35 | 0.002 | 0.05 | 0.005 | 0.13 | 0.010 | 0.25 |
| .250-2.000 | 6.35- 50.8 | .005 | 0.13 | .010 | 0.25 | .015 | 0.38 |
| 2.000-4.000 | 50.8 -101.6 | .010 | 0.25 | .015 | 0.38 | .020 | 0.51 |
| 4.000-6.000 | 101.6 -152.4 | .010 | 0.25 | .015 | 0.38 | .030 | 0.76 |
| Squareness within: | | 0.5 degree per 1 inch length | | 1.5 degrees per 1 inch length | | 2.5 degrees per 1 inch length | |

Parallelism and flatness of ends - See note next page.



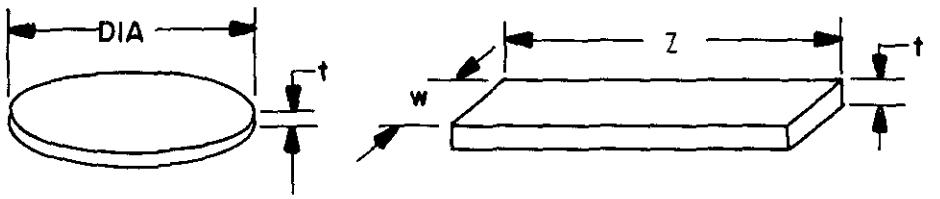
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NOTE: The tolerance for the parallelism and flatness of the ends of the tube is related to the diameter, wall thickness and length. Smaller sizes of diameter and length can be manufactured with tolerances of 0.03 mm (0.001 inch). The larger diameter and lengths in the order of 160 mm (6.30 inches) cannot be held to this tolerance and can be expected to run within 0.25 to 0.30 mm (0.010 to 0.012 inch). The manufacturer of the ceramic element should be consulted to determine the tolerance that can be guaranteed for a particular size ceramic element in production quantities.



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TABLE VI. Mechanical tolerance levels for plates and discs.




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| Disc (dia) Plates (Z & w) Inch | Disc (dia) Plates (Z & w) Millimeter | Level A Fully machined (±) Inch | Level A Fully machined (±) Millimeter | Level B Minimal machining (±) Inch | Level B Minimal machining (±) Millimeter | Level C as fired (±) Inch | Level C as fired (±) Millimeter |
|---|---|---|---|---|---|---|---|
| 0.125-1.500 | 3.2 - 38.1 | 0.003 | ±0.08 | 0.010 | 0.25 | 0.015 | 0.38 |
| 1.500-2.500 | 38.1 - 63.5 | .005 | ±0.13 | .015 | 0.38 | .020 | 0.50 |
| 2.500-3.500 | 63.5 - 88.9 | .005 | ±0.13 | .020 | 0.50 | .025 | 0.64 |
| 3.500-4.500 | 88.9 -114.3 | .010 | ±0.25 | .025 | 0.64 | .040 | 1.02 |
| 4.500-6.000 | 114.3 -152.4 | .010 | ±0.25 | .030 | 0.76 | .050 | 1.27 |
| Disc and plate (thickness) (t) | | | | | | | |
| 0.010-0.015 | 0.25- 0.38 | 0.001 | ±0.03 | 0.002 | 0.05 | 0.002 | 0.05 |
| .015- .035 | .38- .89 | .001 | ±0.03 | .002 | 0.05 | .003 | 0.08 |
| .035- .080 | .89- 2.03 | .002 | ±0.05 | .003 | 0.08 | .004 | 0.10 |
| .080- .200 | 2.03- 5.08 | .003 | ±0.08 | .008 | 0.20 | .010 | 0.25 |
| .200- .500 | 5.08- 12.70 | .004 | ±0.10 | .010 | 0.25 | .015 | 0.38 |
| .500-1.0 | 12.70- 25.40 | .005 | ±0.13 | .020 | 0.50 | .025 | 0.64 |
| Parallel within: | | 0.001 | 0.03 | 0.003 | 0.08 | 0.007 | 0.18 |
| Squareness within: | | 0.75 degrees | | 1.5 degrees | | 2.5 degrees | |
| Flatness: (max diameter for disc 2 inches or plates Z or w 2 inches) | | 0.001 | 0.03 | 0.003 | 0.08 | Within thickness tolerance up to 1 inch diameter and up to 0.080 inch thick. Within 0.005 inch above 0.080 inch thick. | |

NOTE: The flatness of ceramic elements with large diameter-to-thickness ratio is difficult to maintain. When discs or plates with dimensions larger than 50.8 mm (2.0 inches) are required the manufacturer should be consulted on the tolerance that can be maintained on production quantities.



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TABLE VII. Suggested ordering data and specification guide.

1. Application:

Projector _____ Hydrophone _____
Other _____(describe)_____
_____

2. Ceramic type: I_____; II_____; III_____; IV_____; Modified_____

| Property: | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| Nominal value required: | $K_{33}^{T}$ ____ | tan $\delta$ ____ | $k_{eff}^{2}$ ____ | $k_p$ ____ | $N_1$ ____ | $N_p$ ____ | $Q_m$ ____ | Other ____ |

3. Geometric description:

Cylinder_____; Disc_____; Plate____; Bar____; Sphere____; Hemisphere____

Other_____(describe)_____
_____
_____

4. Dimensional requirements:

| | | Tolerance level | | | |
|---|---|---|---|---|---|
| | | A | B | C | *AA |
| O.d. | _____ | | | | |
| Length | _____ | | | | |
| Wall | _____ | | | | |
| I.d. | _____ | | | | |
| Width | _____ | | | | |
| Flatness | _____ | | | | |
| Parallelism | _____ | | | | |
| Perpendicularity (squareness) | _____ | | | | |

*As specified in equipment specifications See reference drawing no._____

5. Other special requirements _____
_____
_____



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TABLE VII. Suggested ordering data and specification guide. - Continued

6. Chips; pits and pocks:

| | Open chips | Closed chips | Pits | Pocks |
|---|---|---|---|---|
| Maximum allowable size | | | | |
| Maximum number per surface | | | | |

Reference drawing number _____

7. Electrode material:

Silver_____; electroless nickel_____

Other_____ (specify)_____

8. Placement of electrodes:

Major surfaces _____; Ends _____; O.d.-I.d._____; Stripes _____

Other_____

Borders_____, Describe_____

_____

9. Electrode connection:

None _____; Wire size _____(describe wire type and stranding)_____

_____

Location of point of attachment_____

Other special requirements_____

10. Poling direction: Radial [ ]; Longitudinal [ ]; Tangential [ ]; Other [ ]

Describe_____

11. Reference drawing number of ceramic element _____



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TABLE VII. Suggested ordering data and specification guide. - Continued

12. Electrical properties (10 days after poling) low field

| | |
|---|---|
| Capacitance | $C^T$ _____ pF(± _____ pF) |
| Dielectric loss factor (max value) | tan δ _____ Not required unless different from value specified in tables II and III. |
| Maximum admittance magnitude 1/ | $Y_m$ _____ siemens (S) (minimum value) |
| Frequency of maximum admittance 1/ | $f_m$ _____ Hz (± _____ Hz) |
| Minimum admittance magnitude 1/ | $Y_n$ _____ siemens (S) (maximum value) |
| Frequency of minimum admittance 1/ | $f_n$ _____ Hz (± _____ Hz) |

1/ These parameters or their equivalent shall be specified for the production ceramic element.

13. High field measurements required: Yes ☐; No ☐

Maximum change in $K_{33}^T$ 0 to 50°C; as specified in table IV ☐

Different value required; _____ percent ☐

Maximum dielectric loss factor; as specified in table IV ☐

Different value required; tan δ _____ ☐

14. Sets of electrical measurements required on production ceramic elements:

One ☐; Two ☐; Time between measurements; _____ days.

15. Test data sheet required: Yes ☐; No ☐

Test data format to be approved by contracting activity ☐

16. Estimated production quantities

Minimum _____ Maximum _____

17. Number (quantity) of first article ceramic elements _____

18. Manufacturers test procedure approval required Yes; _____ No; _____

Approval required by _____
(Activity - Title - Name)



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Material: Type_____ of DOD-STD-1376

Capacitance _____pF ± _____pF (1 kHz)

Maximum admittance $|Y_m|$_____S
(min value)

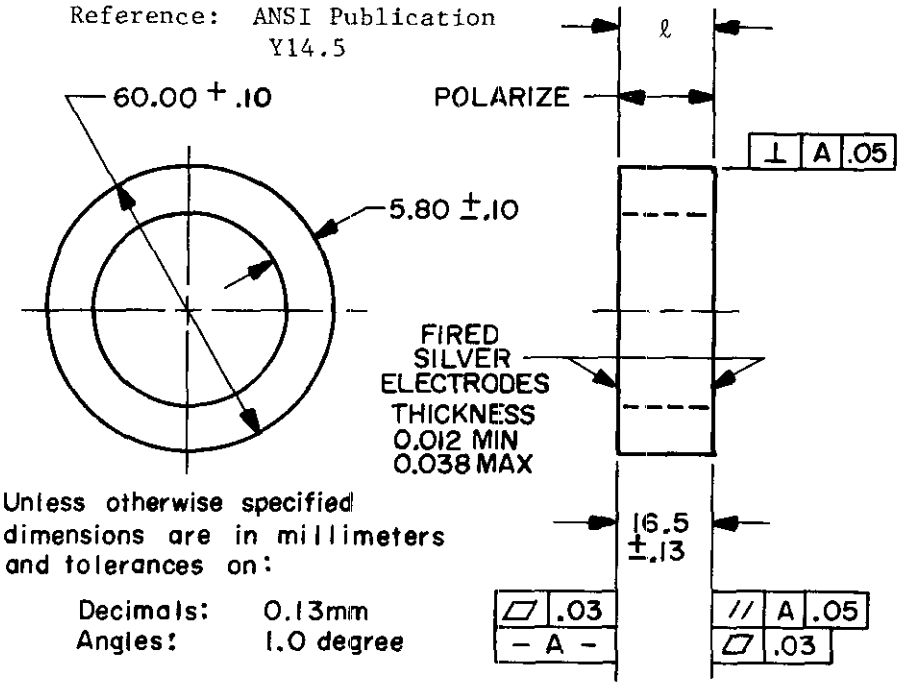
Frequency $f_m$ at $|Y_m|$_____Hz ± _____Hz

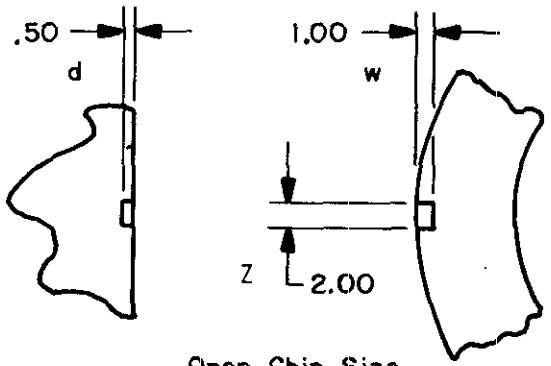
Minimum admittance $|Y_n|$_____S
(max value)

Frequency $f_n$ at $|Y_n|$_____Hz ± _____Hz

Notes:

1. The number of chips or edge irregularities shall not exceed five total. The number of chips on any surface shall not exceed two.
2. Markings indicating the type material shall be color coded in nonconducting ink and shall be placed as close to the positive electrode as practical on the outside cylindrical surface.
3. Mark the poling date in nonconducting ink on the outside cylindrical surface.
4. The ends and outside cylindrical surface finish shall not exceed 1.02 micrometer (40 microinches) before electroding.
5. Electrodes: The plane surface shall be fully covered with fired-on silver electrodes (duPont 6010 or equivalent). The silver shall not extend onto the curved surfaces (inside or outside).




Open Chip Size
Max. value for one to two chips. No other chip shall exceed 0.25mm in d, w, Z

SH 12569

FIGURE 6. Typical format for production ceramic elements - ring.



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*(See Instructions – Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| DOD-STD-1376A(SH) | PIEZOELECTRIC CERAMIC FOR SONAR TRANSDUCERS (HYDROPHONES AND PROJECTORS) |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional |
|---|---|
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